

BARMA-Python: Modeling Relative Humidity in Brasília with the BARMA Model

Everton da Costa

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1 Introduction

This report documents BARMA-Python, a Python implementation of the Beta Autoregressive Moving Average (β ARMA) model for time series bounded in $(0, 1)$. The methodology was originally proposed by Rocha and Cribari-Neto (2009) and is currently available in R via the `betaARMA` package on CRAN, developed by Costa, Cribari-Neto and Scher (2026). This Python port reproduces the package’s core estimation routines and validates them against the R reference to machine precision.

The motivation for porting the methodology to Python is twofold: first, to make β ARMA accessible to the broader Python statistical and machine learning ecosystem; and second, to enable future comparisons of computational performance, numerical stability, and convergence behavior between R and Python implementations. In our recent work on ridge penalization (Cribari-Neto, Costa and Fonseca, 2025), we observed that parameter estimates can be adversely affected by the flatness of the log-likelihood curvature, leading to a significant number of convergence failures. Whether Python’s optimization routines (e.g., `scipy.optimize.minimize`) reduce this convergence failure rate compared to R’s `optim` function remains an open question and a target of future work.

2 Time Series: Relative humidity in Brasília

2.1 Load the time series

The full monthly series spans from January 1999 to the present and was fetched from the NASA POWER API via `fetch_humidity_brasilia.py`. Values are expressed as proportions in $(0, 1)$.

```
PROCESSED_DIR = Path("data/processed")

y_full: pd.Series = pd.read_csv(
    PROCESSED_DIR / "y_full_monthly_data.csv",
    index_col="Month",
    parse_dates=["Month"],
).squeeze("columns")
y_full.name = "y"
y_full.index.freq = pd.infer_freq(y_full.index)

display(y_full)
```

```
Month
1999-01-31    0.821077
```

```

1999-02-28    0.776082
1999-03-31    0.828994
1999-04-30    0.760300
1999-05-31    0.670061
...
2025-12-31    0.773726
2026-01-31    0.755781
2026-02-28    0.791325
2026-03-31    0.770852
2026-04-30    0.670300
Freq: ME, Name: y, Length: 328, dtype: float64

```

We split the data set, using 80% to train the model and 20% to test the accuracy of the predictions.

The train data starts in 1999-01 and ends in 2020-10.
It contains 262 observations.

The test data starts in 2020-11 and ends in 2026-04.
It contains 66 observations.

2.2 Descriptive analysis

Table 1: Descriptive statistics for monthly relative humidity in Brasília, 1999–2026 ($n = 328$).

	count	mean	std	min	25%	50%	75%	max
y	328.0	0.6737	0.1602	0.2932	0.5328	0.7344	0.817	0.8861

The excess of kurtosis is -1.0617 and the skewness is -0.5682, indicating a mildly left-skewed, platykurtic distribution consistent with the bounded time series.

2.3 Seasonality

The seasonal line plot below shows the monthly humidity trajectory for each year, revealing a consistent dry season (June-September) and wet season (November-March).

The box plot below summarizes the within-month variability, confirming the strong seasonal signal and highlighting outlier years.

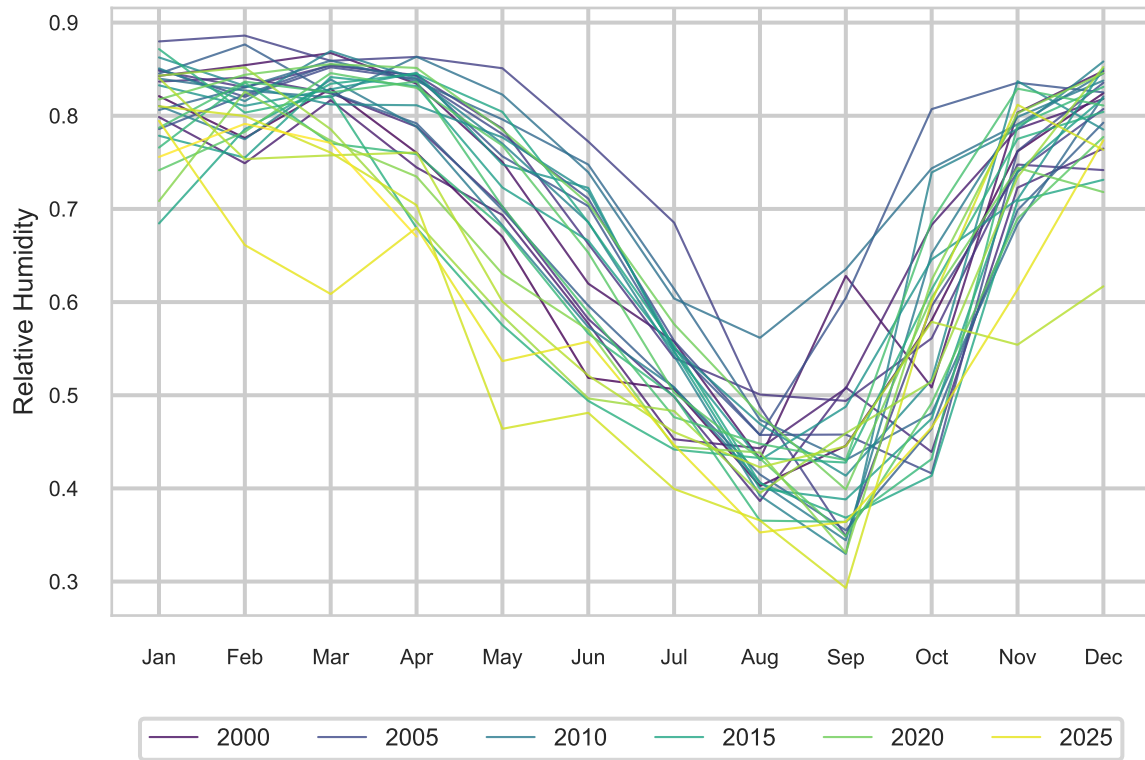


Figure 1: Annual overlays of monthly relative humidity, 1999–2026 (2026 is partial, through April). The plot reveals a strong seasonal cycle with peak humidity during the wet season (November–March) and a pronounced dry season (June–September).

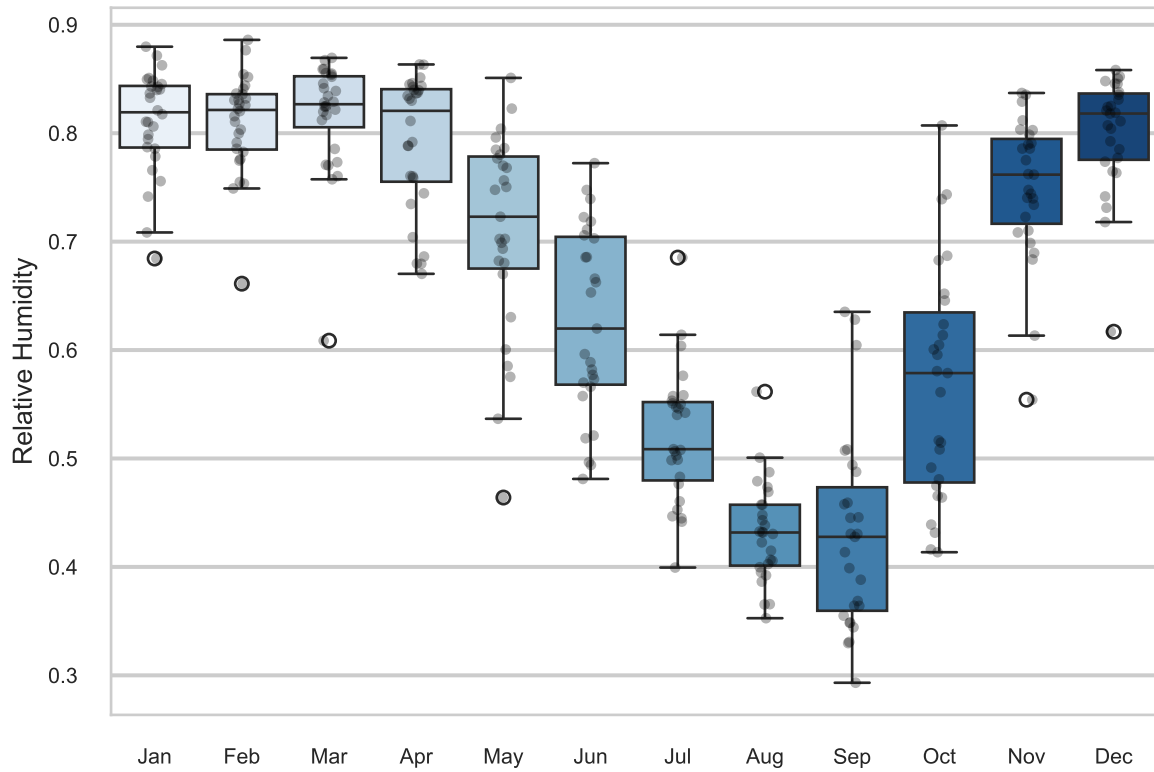


Figure 2: Monthly distribution of relative humidity. Box plots confirm a strong seasonal signal, with the lowest medians in August and September. This cyclical pattern motivates the inclusion of harmonic regressors in the model.

2.4 Harmonic Regressors

In the previous section we demonstrated the presence of strong seasonality presence in the data. To account for the annual seasonal cycle, we include two harmonic regressors, a sine and cosine term with period 12 months.

```
freq = 12
n_train = len(y_train)
n_test = len(y_test)

trend_train = np.arange(1, n_train + 1)
trend_test = np.arange(n_train + 1, n_train + n_test + 1)

phase_train = 2 * np.pi * trend_train / freq
phase_test = 2 * np.pi * trend_test / freq

# harmonic: train
hs_train: np.ndarray = np.sin(phase_train)
hc_train: np.ndarray = np.cos(phase_train)

X_train: np.ndarray = np.column_stack((hs_train, hc_train))
print(f"The shape of the X_train matrix is: {X_train.shape}.")

# harmonic: test
hs_test: np.ndarray = np.sin(phase_test)
hc_test: np.ndarray = np.cos(phase_test)

# X_test carries a date index (unlike X_train) so it aligns with y_test
# when passed to .forecast() / .plot_forecast() later in the report.
X_test: pd.DataFrame = pd.DataFrame(
    np.column_stack((hs_test, hc_test)),
    index=y_test.index
)

print(f"The shape of the X_test matrix is: {X_test.shape}.")
```

The shape of the X_train matrix is: (262, 2).

The shape of the X_test matrix is: (66, 2).

3 β ARMA Model

To align with the approach used in the R package (Costa, Cribari-Neto and Scher, 2026), we are using the response variable in the predictor scale. Additionally, following the authors' recommendation, we will use Pearson residuals in the Residuals Analysis.

3.1 Model Specification

When the seasonality is strong, even harmonic regressors might not be sufficient. In this case, an option provided by the β ARMA models consists of using a specific range of lags during model selection.

For example, we started with the β AR model without an MA component and computed the estimates starting from lag 1 up to 36 to verify the presence of seasonality over a 3-year period; after this, we did it again for the pure β MA models. We consolidated these values into a table and added the estimates of the β ARMA model using only the intercept and the precision estimate, without AR or MA terms. The target was the minimal BIC (the AIC yielded coinciding results for this procedure).

We applied this method for each component we added to the model until we had three finalists. The model with the smallest BIC was the β ARMA model with AR lags at positions 1, 21, 24, and MA at 1 and 26, using the logit link function. The stopping criterion for this procedure was the presence of white noise in the residuals. Once achieved, we stopped the search.

3.2 Fitted Model

We first define the AR and MA components of the β ARMA model following the procedure described in the previous section, along with the logit link function.

```
ar = [1, 21, 24]
ma = [1, 26]
link = "logit"
```

Next, we instantiate the β ARMA model object. We pass the training data, the defined lags, and the matrix of exogenous harmonic regressors defined previously in Section 2.4.

```
barma_model = BARMA(
  y=y_train,
  ar=ar,
  ma=ma,
  exog=X_train,
```

```
link=link
)
```

With the model structure defined, we estimate the parameters using the `.fit()` method.

```
barma_fit = barma_model.fit()
```

We can confirm the optimizer's convergence via the `.converged` attribute:

```
print(f"Converged: {barma_fit.converged}")
```

Converged: True

The model summary can be observed with the `.summary()` method.

```
display(barma_fit.summary())
```

Table 2: Estimated coefficients, standard errors, z-values, and p-values for the β ARMA model: autoregressive terms (φ), moving average terms (θ), harmonic regressors (β), intercept (α), and precision (ϕ). All parameters are significant at the 5% level. AIC and BIC are reported below the table.

	Estimate	Std. Error	z value	Pr(> z)
alpha	0.598980	0.109690	5.460655	4.743811e-08
varphi1	0.310446	0.095389	3.254531	1.135799e-03
varphi21	-0.209811	0.055167	-3.803170	1.428563e-04
varphi24	0.207745	0.057058	3.640925	2.716599e-04
theta1	0.256996	0.104479	2.459773	1.390250e-02
theta26	-0.162686	0.068925	-2.360322	1.825909e-02
beta1	0.860752	0.048161	17.872319	0.000000e+00
beta2	0.399676	0.046551	8.585711	0.000000e+00
phi	64.147535	5.871704	10.924858	0.000000e+00

The AIC, BIC values can be retrieved using the `.aic` and `.bic` attributes, respectively.

```
print(f"AIC: {barma_fit.aic:.4f}.")
print(f"BIC: {barma_fit.bic:.4f}.")
```

AIC: -692.8546.

BIC: -660.7395.

3.2.1 Residual analysis

Using the estimates of the model obtained in the previous section, we can proceed with the residual analysis.

Some plots can be very helpful in this fundamental part. We can access some diagnostic plots using the `.plot_diagnostics()` method.

In the following figure, we can see four plots. In the Observed vs fitted panel, we can see a very good alignment between the model predictions and the observed values. In the Residuals over time plot, we include a ± 3 band. Because the residuals do not follow a normal distribution, we use an ad hoc recommendation provided by the authors. More details about this can be found in (Rocha and Cribari-Neto, 2009). In this plot, one value (December 2011) falls slightly outside the band. Two very important plots to examine are the Residual ACF and Residual PACF; in these plots, we can see that no values fall outside the confidence bands.

With the conclusions presented in the figure above, we have a strong indication of white noise in the residuals. To confirm that, we could execute the Ljung-Box test. To do that, we can use the method `.plot_ljungbox()`.

In the Ljung-Box p-values plot, we can see the lags starting from 6 until 24. The lower bound is the number of AR and MA parameters plus one ($3 + 2 + 1 = 6$), and the upper bound covers two full annual cycles. All p-values are greater than 0.05, meaning we do not reject the null hypothesis of white noise residuals.

Therefore, based on both the visual plot analysis and the statistical test p-values, we conclude that the residuals behave as white noise. Consequently, we can confidently trust the inferences drawn using the model's estimates.

Therefore, the forecast plot can be accessed by the `.plot_forecast()` method, using `y_test` to check prediction alignment and `exog` matrix for prediction.

4 Baseline Models

Now, we evaluate the accuracy of the forecasts using the β ARMA model, following the model selection and evaluation procedure of Costa, Cribari-Neto and Scher (2024). We reserved 20% of the dataset (66 observations, roughly 5.5 years) for the test set. Forecast accuracy is evaluated using MAE and RMSE; the non-seasonal and seasonal integration orders are set to $d = D = 0$ since the response is bounded in $(0, 1)$.

We consider traditional forecast strategies to benchmark the forecast from the β ARMA model. Specifically, we use SARIMA (seasonal autoregressive integrated moving average) with $s = 12$ because we have monthly data; and DHR (Dynamic Harmonic Regressors) (Young, Pedregal and Tych, 1999; Hyndman and Athanasopoulos, 2021), using the same harmonic regressors as

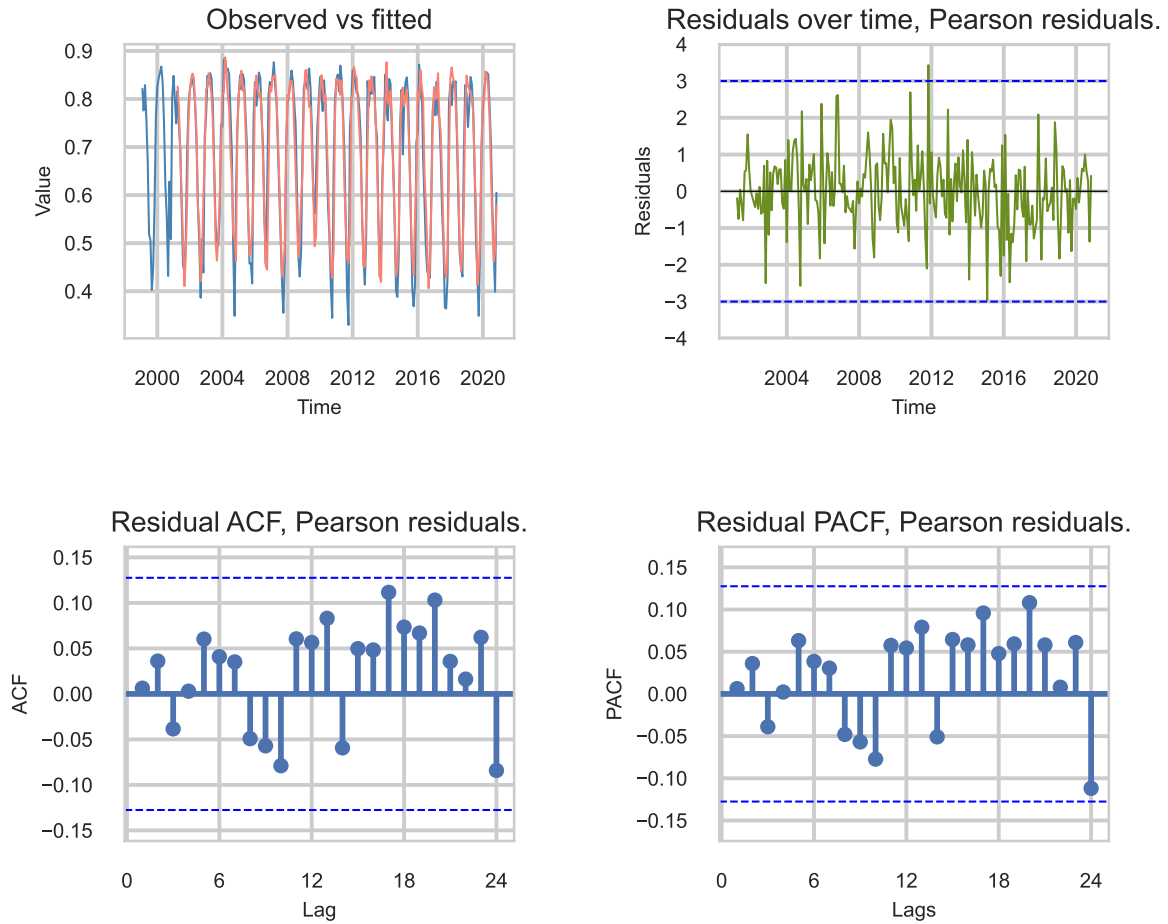


Figure 3: Four-panel residual diagnostics: observed vs. fitted values, Pearson residuals over time (with ± 3 reference bands), residual ACF, and residual PACF. Residuals center around zero, and no ACF or PACF spike exceeds the confidence bounds — consistent with the model having captured the temporal dependence.

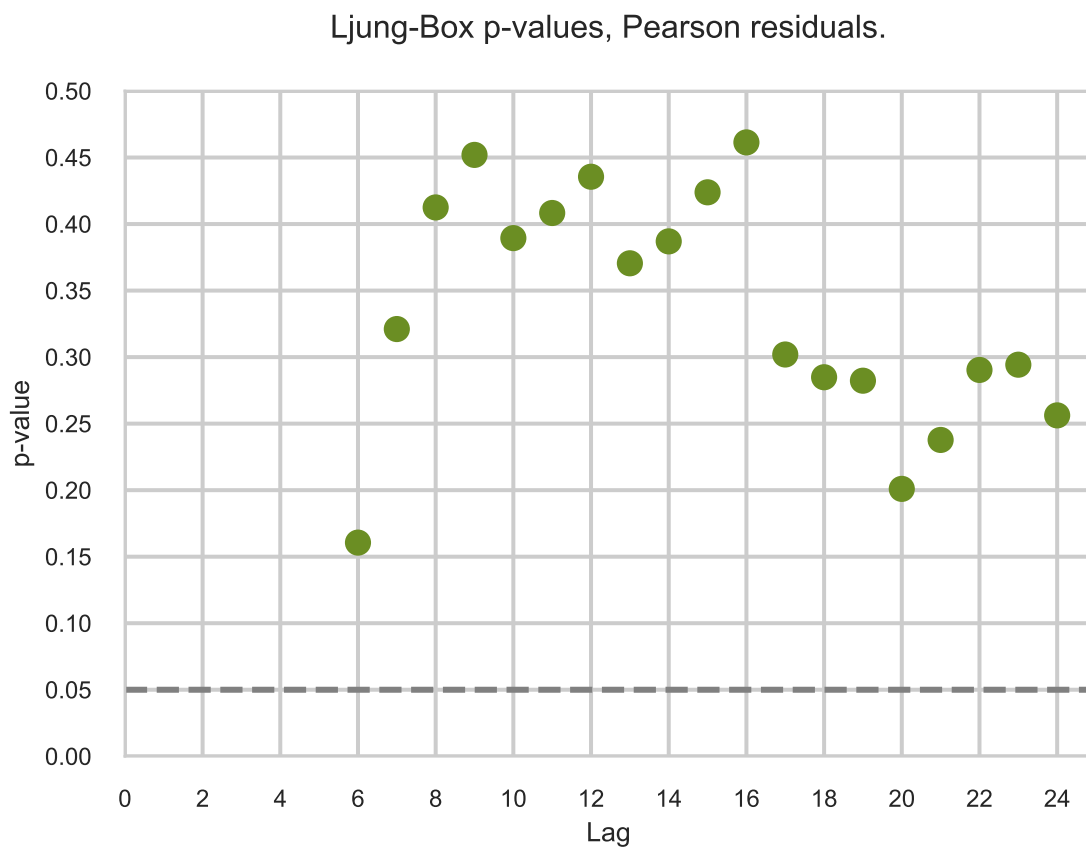


Figure 4: Ljung–Box test for residual autocorrelation. P-values for the Pearson residuals at lags 6–24 all lie above the 0.05 threshold (dashed line), so the null hypothesis of no residual autocorrelation is not rejected. The result supports the adequacy of the β ARMA specification.

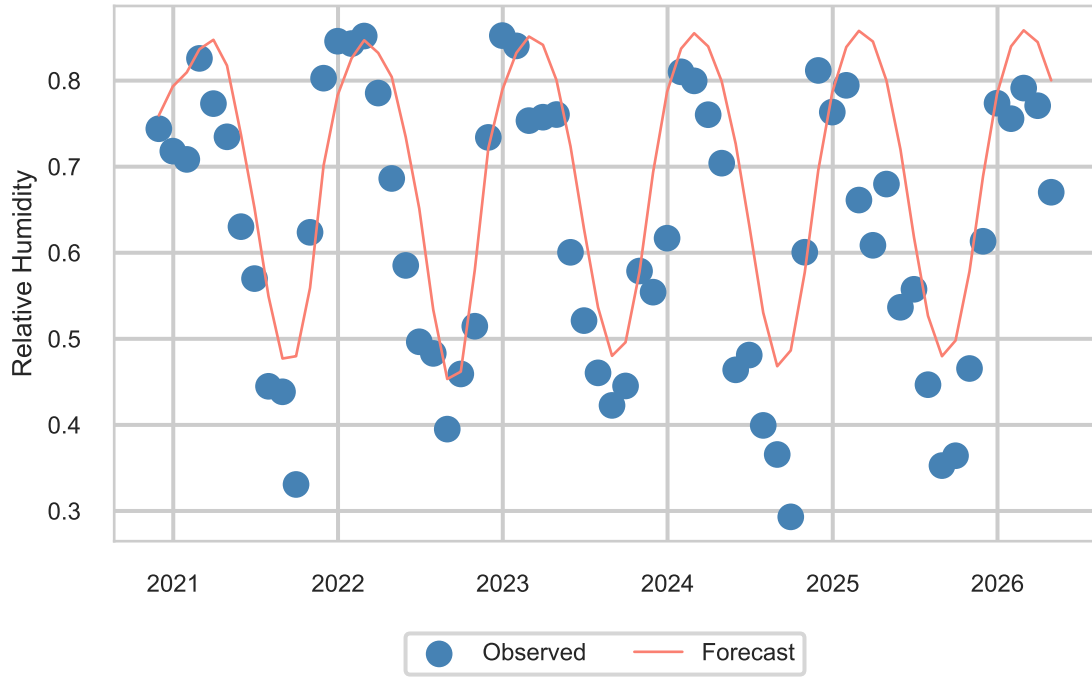


Figure 5: Out-of-sample β ARMA forecast (solid line) against observed relative humidity (points) over the 66-month test window. Predictions track the seasonal cycle and remain within the valid $(0, 1)$ range by construction, a guarantee ARIMA-family models do not provide.

the β ARMA model (sine and cosine). To verify the impact of the seasonality in the model, we also use an ARIMA model without any seasonal component or harmonic regressor.

In the three baseline models, we set the degree of differencing to zero and use the function `auto_arima()` from the `pmdarima` package. In the next three code chunks, you can see the code used for each model.

```
# Approach 1: DHR
dhr_fit = auto_arima(
    y=y_train,
    X=X_train,
    d=0, D=0,
    with_intercept=True,
    trend=None,
    seasonal=None
)

# display(dhr_fit.summary())

y_hat_dhr_fit = dhr_fit.predict(n_periods=len(y_test), X=X_test)
rmse_dhr = np.sqrt(np.mean((y_test.values - y_hat_dhr_fit.values) ** 2))
mae_dhr = np.mean(np.abs(y_test.values - y_hat_dhr_fit.values))

plot_dhr_fit = plot_diagnostics(
    y=y_train, fitted_values=y_hat_dhr_fit, residuals=dhr_fit.resid(),
)

# display(plot_dhr_fit)
plt.close(plot_dhr_fit)
```

```
# Approach 2: SARIMA
sarima_fit = auto_arima(
    y=y_train,
    d=0, D=0,
    with_intercept=True,
    trend=None,
    seasonal=True,
    m=12
)

# display(sarima_fit.summary())
```

```

# SARIMA predictions
y_hat_sarima_fit = sarima_fit.predict(n_periods=len(y_test))
rmse_sarima = np.sqrt(np.mean((y_test.values - y_hat_sarima_fit.values) ** 2))
mae_sarima = np.mean(np.abs(y_test.values - y_hat_sarima_fit.values))

# SARIMA residuals
plot_sarima_fit = plot_diagnostics(
    y=y_train, fitted_values=y_hat_sarima_fit, residuals=sarima_fit.resid()
)
# display(plot_sarima_fit)
plt.close(plot_sarima_fit)

```

```

# Approach 1: ARIMA
arima_fit = auto_arima(
    y=y_train,
    d=0, D=0,
    with_intercept=True,
    trend=None,
    seasonal=False
)
# display(arima_fit.summary())

# ARIMA residuals
y_hat_arima_fit = arima_fit.predict(n_periods=len(y_test))
rmse_arima = np.sqrt(np.mean((y_test.values - y_hat_arima_fit.values) ** 2))
mae_arima = np.mean(np.abs(y_test.values - y_hat_arima_fit.values))

# ARIMA residuals
plot_arima_fit = plot_diagnostics(
    y=y_train, fitted_values=y_hat_arima_fit, residuals=arima_fit.resid()
)

# display(plot_arima_fit)
plt.close(plot_arima_fit)

```

5 Results

In Table 3, we have the MAEs and RMSEs (x100) of the β ARMA forecast together with DHR (order : (2, 0, 3)), SARIMA (order : (1, 0, 0)(1, 0, 0)₁₂) and ARIMA (order : (2, 0, 3)) models.

The Ljung–Box portmanteau tests do not show evidence of DHR, SARIMA or ARIMA misspecification at the usual significance levels.

We can note that seasonality is strong in the dataset and the model must deal with it. The ARIMA model yielded the highest errors among the 4 forecast models. The SARIMA model provided an improvement when compared to the ARIMA model, but remains distant from the two other models. β ARMA outperforms DHR and achieves the best results among the 4 forecast models. It is important to note that the β ARMA and DHR predictions are very competitive, but in the structure of the β ARMA model we can always be sure that the predictions and the fitted values stay in the interval $(0, 1)$. We do not have this guarantee with the other three models. For readers interested in seeing a full comparison between the models, we invite them to read (Costa, Cribari-Neto and Scher, 2024).

Table 3: Forecast accuracy on the 66-month test set. MAE and RMSE are reported on the $(0, 1)$ scale, multiplied by 100 for readability. β ARMA achieves the lowest error on both metrics.

	Model	MAE (%)	RMSE (%)
1	BARMA	8.73	10.39
2	DHR	8.90	10.57
3	SARIMA	11.22	13.52
4	ARIMA	11.88	14.38

In addition to the table results, we also have Figure 6. In this figure, we can see observed values as points and the forecasts of each model as lines. The degradation of the predictions from the ARIMA and SARIMA models over time is notable. Also notable is the close competition between the β ARMA and DHR models, sometimes resulting in the overlapping of the lines.

6 Conclusion

In this work, we successfully imported the main functionalities of the β ARMA model from the R package (Costa, Cribari-Neto and Scher, 2026) into Python. Backed by unit tests via `pytest`, we presented the initial version of a future Python package. We demonstrated these functionalities using a real-world dataset of relative humidity in Brasilia, Brazil. The report provides not only the model fitting process, but also the evaluation of predictions and residuals using ACF, PACF and Ljung-Box tests for lags 6 through 24.

We also benchmarked our implementation against DHR, SARIMA, and ARIMA approaches using the `pmdarima` package. Evaluating the MAE, RMSE, and the residual diagnostics made the advantage of the β ARMA model clearly evident: it yielded white noise residuals, and error metrics that were highly competitive, or superior, to the traditional baselines.

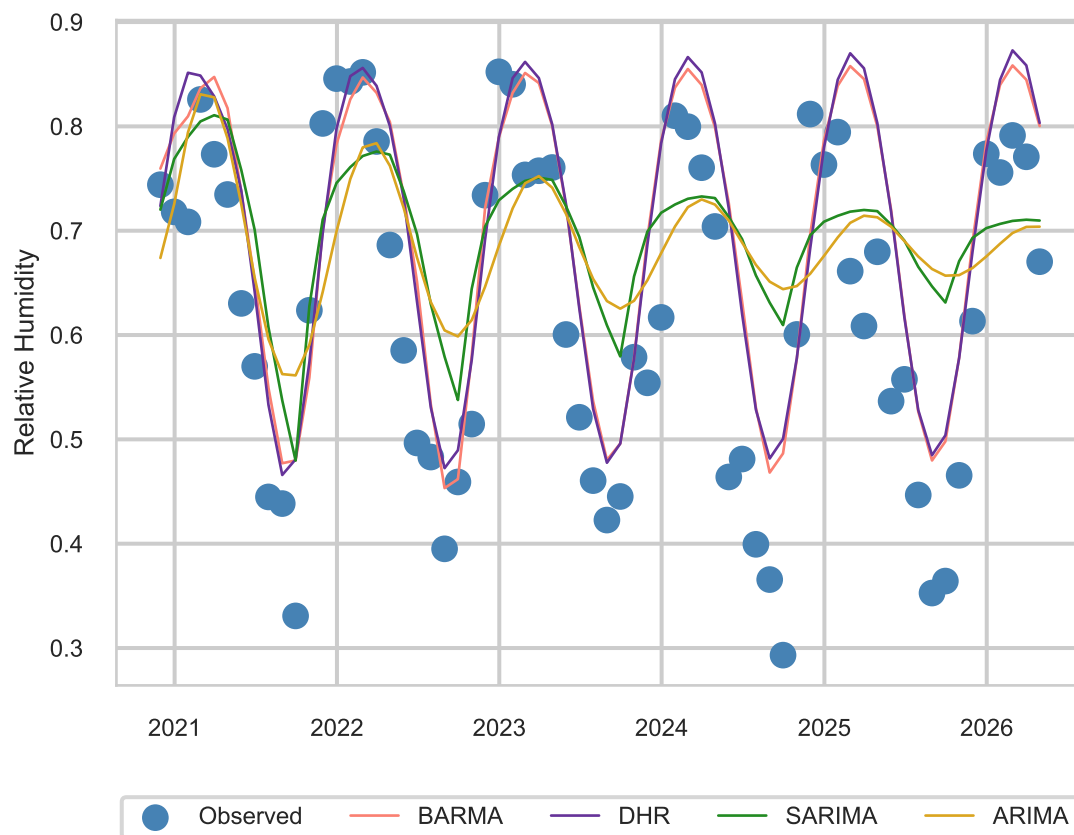


Figure 6: Out-of-sample forecast comparison across the four models. Visually, β ARMA and DHR track the seasonal cycle closely, while SARIMA and ARIMA collapse toward the mean of the training set.

For the sake of brevity, the full comparison was not the primary target of this report; rather our goal was to showcase the packages’s functionality and practical application with the Python ecosystem.

For readers seeking a deeper dive into theoretical and practical advantages of this methodology, we invite you to read our publication in the *Journal of Hydrology* (Costa, Cribari-Neto and Scher, 2024). In that paper we provide a comprehensive comparative analysis exploring the full advantages of β ARMA model. Specifically, we modeled the useful volume proportion of three distinct water reservoirs in the São Francisco River basin, a critical water source for Brazil’s northeast region. The paper highlights crucial theoretical guarantees, such as the fact that the β ARMA model inherently restricts fitted and predicted values to the strictly valid $(0,1)$ interval, a property not guaranteed by ARIMA-family approaches.

7 Future works

We successfully achieved the first goal in this project, which was to make the β ARMA model accessible to the broader Python statistical and machine learning ecosystem. Due to time constraints and the extensive scope of this work, we will defer questions regarding computational performance and numerical stability to future work. The computationally intensive simulation studies required for this analysis will be conducted in the next phase.

References

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